

Andrew Zheng

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EXPERIENCE

Software Engineer Intern | [Miovision](#) | Waterloo, ON

May 2026 – Aug 2026

- Re-engineered RAUC firmware update path for Yocto-based x86-64 Linux devices in Node.js and AWS IoT via adaptive block hash index HTTP streaming to reduce OTA bandwidth cost by **98%** (1.4GB to 21MB)
- Built end-to-end HIL release pipeline for embedded ARM64 Linux devices in Node.js and AWS/GitLab on Terraform-provisioned, Ansible-configured OTA runners to poll **15** health metrics over **48h** with **<1%** frame-drop
- Deployed 360° camera emulator app to embedded ARM64 Linux devices on Firecracker microVM in Go/Docker to virtualize H.264 RTSP streaming + camera API, enabling camera-free HIL development
- Architected fleet-wide dual lookup for device search engine in Kotlin/Spring Boot to stream schema registry Kafka data to Kubernetes-hosted index, eliminating **100%** of stale results across **43K+** devices

Embedded Systems Developer | [Waterloo Rocketry](#) | Waterloo, ON

Sep 2025 – Present

- Designed 4-layer recovery parachute PCB for the **world record highest amateur liquid rocket (63,497 ft)**, integrating dual 9V LiPo power stages with StratoLogger altimeters and Rocket CAN bus
- Engineered hardware fail-safe for PIC18 microcontroller to drive a P/N-channel MOSFET network to arm dual altimeters, fire pyros, and guarantee parachute deployment at peak altitude

Founder & Case Coach | [Case Tutoring Startup](#) | Toronto, ON

May 2024 – Aug 2025

- Drove **\$10K+** revenue through GTM campaign for self-founded business case contest coaching service, scaling across **150+** high school students from **10+** schools
- Won **1st Place** globally and coached **50+** qualifiers/**30+** international finalists at DECA ICDC, America's largest high school business case contest (230K+ members)

PROJECTS

Allocate | [macOS CPU/Thermal Fleet Controller](#) | *Rust, XNU, Mach IPC, Prometheus, Grafana*

[GitHub](#) 

- Built system-level macOS CPU governor using XNU QoS and POSIX policies to steer workloads across P/E cores and reduce background CPU consumption by **79%** with **100%** app uptime
- Engineered Rust FFI telemetry daemon to sample CPU, AppleSMC, IOReport, DVFS, and SoC power metrics every **500ms**, displayed on native AppKit live CPU chart
- Built 18-sensor Rust PID-based thermal controller reading CPU/GPU/ANE to dynamically adjust E-core policies through Mach/XPC, reducing CPU temperature by **12°C/22°F** without hardware fans
- Shipped native end-to-end SwiftUI app and Prometheus/Grafana fleet observability pipeline with configurable thresholds for per-app CPU controls, thermal, power, and throttling analysis

GitIRL Bot | [Hack the North 2026 Winner](#) | *Python, FastAPI, SLAM, OpenAI API*

[Devpost](#) 

- Won Best use of Bracket Bot Hardware at Canada's Biggest Hackathon by building an autonomous version control system to map world states and command a humanoid robot to restore environment changes
- Designed fleet orchestration pipeline across Bracket Bot and cloud platform, indexing **6.9M+** telemetry points in Elasticsearch to enable hybrid semantic search processing for robotic actions

CrawlStars | [Distributed Search Engine](#) | *Golang, JavaScript, MongoDB, REST API*

[GitHub](#) 

- Engineered concurrent crawlers in producer-consumer architecture to process **2000+ pages/min** in **<15MB**
- Optimized deduplication to enable O(1) lookup for **50K+** URLs and filter **70%** of redundant crawls
- Deployed search engine to calculate SEO relevance via MongoDB Atlas aggregation pipeline for **50K+** URLs

SKILLS

Languages: C, C++, Python, Rust, Go, Java, Kotlin, Swift, JavaScript, TypeScript, HTML, CSS, Ruby, Verilog

Infrastructure: AWS, Linux, Docker, Kubernetes, Terraform, Ansible, Kafka, CI/CD, Git, Prometheus, Grafana

Backend: Node.js, Express.js, FastAPI, Spring Boot, Ruby on Rails, REST API, MongoDB, Jupyter

Frontend/Other: React, Next.js, Tailwind, Streamlit, XGBoost, SLAM, VLA, Ubuntu, Mach, Apple IOKit

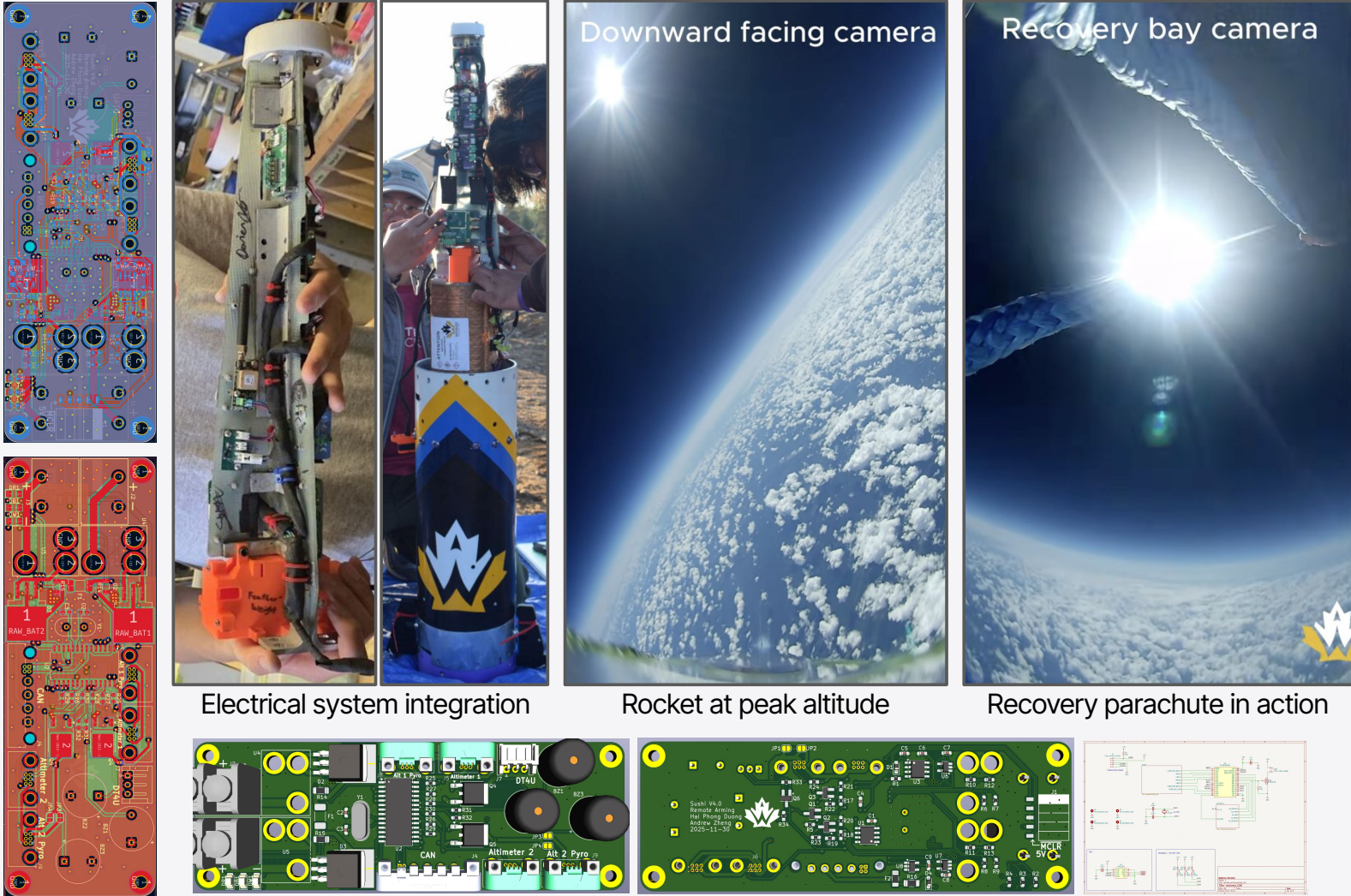
EDUCATION

University of Waterloo | *BASc, Computer Engineering*

Sep 2025 – Apr 2030

World Record Highest Amateur Liquid Rocket (63,497 ft)

Hardware: Safety-Critical KiCad PCB | Dual Power Input | P-MOS High-Side Mag Switch
Firmware: Embedded C (PIC18) | CAN Bus Telemetry Aggregation | ADC Monitoring



GitIRL Bot | Version Control for the Physical World [Devpost]

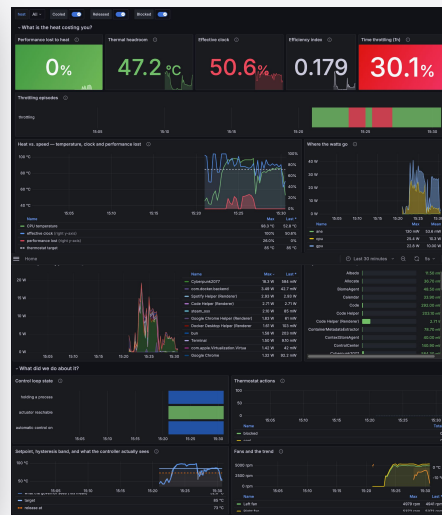
Hack the North 2026 Winner - Best use of Bracket Bot hardware
Python, FastAPI, React/Javascript, Docker, ESQ, OpenAI API | Humanoid VLA with 4D Memory



Rust/C FFI, XNU QoS, Mach IPC | 79% Reduced Background CPU Usage | 100% App Uptime

Albedo | Fleet Thermal Control Agent [[GitHub](#)]

AppleSMC, IOKit | Achieved 12°C/22°F Fanless CPU Cooling | 18 Local Hardware Sensors

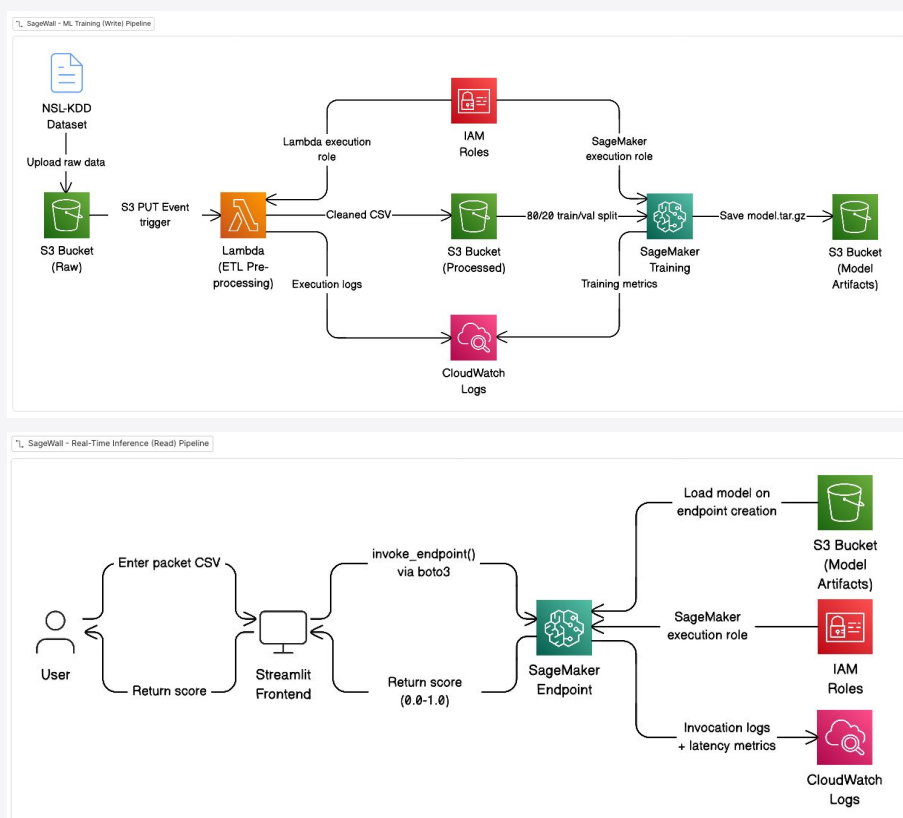


The screenshot displays the Fleet Control app dashboard with the following components:

- Top Navigation:** Includes a search bar, a breadcrumb trail (Home > Dashboards > Agents > Fleet control app), and user information (John D., 10:40 AM).
- Dashboard Header:** Features a 'Filter' button, a 'Last hour' time range selector, and a 'Full screen' icon.
- Row 1 Metrics:**
 - Fleet health**: A green card showing the value '2'.
 - Agents breeding time**: An orange card showing the value '1'.
 - Mean performance loss**: A yellow card showing the value '8.41%'.
 - Peak power draw**: A grey card showing the value '51.4 w'.
 - Least efficient host**: A black card showing values '0.033' and '0.038' for hosts 'lab-2' and 'macbook'.
 - Hosts able to act**: A black card showing the value '1'.
- Row 2 Metrics:**
 - Per host**: A section header for the following charts.
 - Thrashing by host**: A green bar chart showing thrashing levels across hosts.
 - Performance test by host**: A yellow bar chart showing performance test results across hosts.
- Row 3 Metrics:**
 - CPU temperature by host**: A green line chart showing CPU temperature over time.
 - Effective clock by host**: A yellow bar chart showing effective clock rates across hosts.
 - Power by host**: A yellow line chart showing power consumption over time.
- Row 4 Metrics:**
 - Collector status per host**: A yellow bar chart showing collector status across hosts.

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graph TD
    MacUser[Mac User] -- "View and control" --> SwiftUI[SwiftUI Menu Bar]
    MacUser -- "Inspect and govern" --> Ratatui[Ratatui CLI]
    AppleSMC[AppleSMC / IOKit] -- "Sensor readings" --> RustEngine[Rust Thermal Engine]
    IORReport[IORReport / libproc] -- "Power and processes" --> RustEngine
    SwiftUI -- "C ABI snapshots" --> RustEngine
    Ratatui -- "Native Rust calls" --> RustEngine
    RustEngine -- "Expose derived metrics" --> MetricsAPI[Metrics HTTP API]
    RustEngine -- "Request E-core hold" --> XPCDaemon[Allocate XPC Daemon]
    MetricsAPI -- "Scrape host metrics" --> GrafanaAlloy[Grafana Alloy]
    MetricsAPI -- "Local scrape" --> PrometheusTSDB[(Prometheus TSDB)]
    GrafanaAlloy -- "Remote write" --> PrometheusTSDB
    XPCDaemon -- "Apply QoS policy" --> AppleScheduler[Apple Silicon Scheduler]
    PrometheusTSDB -- "Fleet visualization" --> GrafanaDashboards[Grafana Dashboards]
  
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Golang, MongoDB | O(1) lookup time | 2000+ URLs/minute | <15MB memory usage

